Advanced RAG - Course Outline

Pre-requisite: LangGraph & LangChain Basics

Module 1: RAG Fundamentals and Architecture

- Introduction to RAG (What it is and the problems it solves like hallucinations and knowledge cutoffs)
- Core components and RAG data flow (Knowledge Base → Retriever → Generator)
- Comparison: RAG vs Fine-Tuning vs Prompt Engineering

Module 2: Document Processing and Chunking

- Document Loaders in LangChain (PDFs, Web, Structured data)
- Text Splitting Strategies (Recursive, Character, Semantic, Markdown, and Code splitters)
- Chunking Best Practices (Optimal chunk sizes and overlap strategies)
- Metadata Management and filtering

Module 3: Embeddings and Vector Representations

- How Embeddings Work (Vector representations, semantic similarity, and distance metrics)
- Embedding Models Overview (OpenAI vs Open-source alternatives)
- Choosing the Right Embedding Model (Cost vs quality trade-offs)
- LangChain Embeddings Implementation

Module 4: Vector Stores

- Vector Store Working and Indexing Strategies (IVF, HNSW)
- Vector Store Operations (Create, Read, Update, Delete)

Module 5: Basic Retrieval Techniques

- Similarity Search Fundamentals
- Similarity Score Thresholds
- Maximal Marginal Relevance (MMR) to balance relevance and diversity
- Hybrid Search (Combining dense semantic vectors + sparse BM25 vectors)
- Ensemble Retriever

Module 6: Advanced Retrieval Techniques

- Contextual Compression
- Parent Document Retriever
- Self-Query
- Multi-Query Retrievers

Module 7: Advanced RAG Patterns

- RAG Fusion and RRF
- HyDE (Hypothetical Document Embeddings)
- Corrective RAG (CRAG)
- Self-RAG,
- Graph RAG
- Multi Modal RAG

Module 8: Agentic RAG with LangGraph

- Introduction to Agentic RAG and its capabilities
- Using RAG as a Tool for Agents
- LangGraph Fundamentals for RAG (State, nodes, edges, checkpointing)
- Agentic RAG Design Patterns (ReAct, Plan-and-execute, Reflection)

Module 9: RAG evaluation through RAGAS

- What is the RAGAS framework?
- Parts of RAG pipeline to evaluate
- RAG metrics in detail
- Code implementation of individual metrics.
- Code implementation of RAG pipeline using evaluate() API

Module 10 : Capstone Project with Deployment (added very soon)

- Project Assignment: Build a production-ready RAG system
- Multi-document ingestion and hybrid retrieval with re-ranking
- Agentic workflow with LangGraph
- Evaluation suite (RAGAS) and Monitoring (LangSmith)
- System Deployment

Production RAG

- Pipeline Optimization
- Caching Strategies
- Cost Optimization
- Monitoring and Debugging
- Common Pitfalls and Solutions
- Security and Compliance